

UmbreWatch: An Edge-Fog-Cloud IoT Framework for Smart Umbrella Rack Management and Anti-Theft Validation

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Abstract—Customers frequently forget their umbrellas at store entrances when the weather improves and rain stops. Additionally, without proper monitoring, unattended umbrellas are susceptible to theft. To address these challenges, we propose UmbreWatch, a comprehensive Internet of Things (IoT) solution that provides smart tracking, instant alerts, and security for shared umbrella spaces. Utilizing an Edge-Fog-Cloud architecture, the system combines ultrasonic, PIR motion sensor, buzzer and LCD at the edge to track umbrella slot occupancy and customer departures. The fog layer handles facial recognition to validate the identity of users retrieving umbrellas, while the cloud layer leverages InfluxDB and Grafana for city-wide data analytics, alongside Azure OpenAI for image reasoning. Through real-time monitoring and alert mechanisms, UmbreWatch successfully eliminates the inconvenience of lost personal property while providing store owners with actionable foot-traffic insights.

Index Terms—Smart city, AIoT, Edge-Fog-Cloud architecture, theft detection

I. INTRODUCTION

A common real-world pain point for customers is placing umbrellas at a store or restaurant entrance, only to forget them upon departure when the rain stops. This oversight leads to customer inconvenience and the potential loss of personal property. Furthermore, without an active monitoring system, shared umbrellas can easily be taken by unauthorized individuals. The primary goal of this project is to develop “UmbreWatch,” a smart umbrella rack system that prevents people from forgetting their umbrellas while simultaneously providing owners with actionable data analytics to optimize operational efficiency. To achieve these goals, our group contributed a multi-tiered IoT architecture that integrates hardware sensing, local message brokering, machine learning-based facial recognition, and Large Language Model (LLM) reasoning. By combining edge-based alerts with fog-based security and cloud-based analytics, we ensure that users and their umbrellas are never parted.

II. RELATED WORK OR CONCEPTS

Recent advancements in smart city infrastructure have introduced various shared umbrella management systems [1].

However, the majority of existing solutions—such as commercial umbrella rental kiosks—rely heavily on active user registration, QR code scanning, or RFID tagging to track usage [2]. While effective for paid rental models, these systems impose a significant barrier to entry for everyday consumers in retail or dining environments who simply need a secure, temporary space to store their personal umbrellas without downloading an application.

In the realm of IoT occupancy and presence detection, several technical approaches have been explored in recent literature [3]. Weight detection via load cells or pressure mats is a frequently implemented, cost-effective method that determines object presence by monitoring aggregate weight fluctuations on a rack [4]. While this simplifies the hardware architecture by relying on a single central sensor, it fundamentally lacks the per-slot granularity required for individual item tracking [4]. Conversely, ultrasonic and infrared sensing arrays require a higher quantity of hardware components—typically one sensor per slot—but provide precise, localized detection [3]. For advanced IoT management and anti-theft systems, this per-slot data is essential for triggering targeted notifications and compiling detailed, reliable analytics [3].

Furthermore, traditional anti-theft mechanisms in IoT deployments typically rely on centralized cloud architectures for video surveillance and biometric processing [5]. Transmitting continuous high-definition video streams to the cloud not only incurs significant bandwidth costs and introduces network latency, but also raises severe user privacy concerns [6]. Consequently, recent paradigms emphasize the transition towards Edge-Fog-Cloud continuums [5], [7]. By offloading sensitive biometric processing (such as facial recognition) to the local fog layer, systems can achieve “zero-registration” identity verification [6], [7]. This approach preserves user privacy by ensuring biometric data never leaves the local network, an architectural concept that UmbreWatch heavily leverages to secure personal property without relying on centralized identity databases [6].

III. PROPOSED IOT SOLUTION

The proposed UmbreWatch solution employs a robust Edge-Fog-Cloud continuum to manage workflow, messaging, and intelligence. **Edge Layer (Sensing and Alerts):** The physical rack is equipped with a Raspberry Pi paired with a Grove Pi HAT. Grove Ultrasonic Sensors monitor individual slots; when the measured distance is less than 4 cm, the slot is classified as occupied. Two PIR motion sensors monitor the exit doorway. If a customer attempts to leave (triggering the inside PIR sensor first) but the umbrella slot status remains unchanged, a Grove Buzzer sounds an alarm to remind the customer. Additionally, a DHT11 sensor collects temperature and humidity data. **Fog Layer (Messaging and Vision Models):** A local Mosquitto MQTT broker is deployed on a PC to manage local network telemetry. The fog layer operates a "zero-registration anti-theft brain". When an umbrella is placed or removed, the Raspberry Pi camera captures an image. A Python script utilizing DeepFace performs facial recognition locally to ensure a match between the depositor and the retriever. If a mismatch occurs, an alarm system is triggered. **Cloud Layer (Reasoning and Dashboards):** Telemetry is published to InfluxDB, serving as the main data source. Grafana is used to create dashboards displaying slot status, weather conditions, and data analysis. Furthermore, Azure OpenAI is integrated to explain the contents of captured pictures. It can use to describe if the person looks nervous or not. The system also combine OpenWeather API, allowing users get weather data of their district. Node-RED is utilized throughout to program main functionalities and structure the flow logic.

IV. INTELLIGENCE PLACEMENT

The architecture of UmbreWatch is founded upon a strategic distribution of computational intelligence across the Edge-Fog-Cloud continuum to balance latency, privacy, and processing efficiency. By decentralizing logic, time-critical tasks are handled at the Edge, where basic functions—such as distance thresholding at the 4 cm mark and immediate image capture—are executed. This localized approach ensures zero-latency responses for LED indicators and camera triggers the moment a customer interacts with the rack. Transitioning to the Fog layer, the system manages heavy image processing and facial recognition locally to ensure that depositor and retriever matching occurs without external data transmission. This architectural decision mitigates high bandwidth costs and protects user privacy by keeping sensitive biometric data within the local network. Finally, the Cloud layer serves as the repository for long-term data storage and the engine for advanced Large Language Model (LLM) reasoning. By utilizing Azure OpenAI, the system offloads the heavy computational burden of image explanation and natural language processing while aggregating weather data to provide deep analytics and predictive insights.

V. DECISIONS AND OBSERVATIONS

In the pursuit of an optimal presence detection mechanism, the project evaluated two primary technical architectures: Ul-

trasonic Sensing and Weight Detection via Load Cells. While the load cell approach offered a cost-effective simplification by monitoring total weight fluctuations through a single sensor, it ultimately lacked the granularity required for individual slot management. Consequently, the team opted for ultrasonic sensors, which, despite requiring a higher component count, provided the precise per-slot detection necessary for localized tracking. The implementation and debugging phase yielded several critical observations that necessitated a iterative redesign of the hardware layer. Initial testing revealed that ultrasonic sensors frequently produced inaccurate distance readings due to physical obstacles and acoustic reflections within the compact rack structure. To resolve this, the physical positioning of the sensors was iteratively adjusted to ensure a dedicated, clear line of sight for each umbrella slot. Furthermore, a significant compatibility issue involving the Grove Pi operating system version caused the PIR motion sensors to malfunction on digital ports. This constraint forced a structural transition to analog ports, which successfully stabilized the motion detection capabilities required for the departure alert system. Software architecture decisions were equally influenced by external service limitations and privacy considerations. The team originally intended to utilize the Azure Face Recognition service; however, stringent safety and privacy policies restrict this tool to corporate entities with specific approval. This limitation prompted a pivot to open-source alternatives. After evaluating several machine learning libraries, DeepFace was selected as the engine for the "zero-registration anti-theft" feature. DeepFace demonstrated superior stability and accuracy in verifying identity matches between depositors and retrievers, providing a robust security layer without the complexity of a centralized user database.

VI. CONCLUSION

The UmbreWatch project successfully demonstrates a highly capable, multi-tier IoT system that solves the practical issues of forgotten and stolen umbrellas. By employing an Edge-Fog-Cloud architecture, the system prioritizes user privacy by retaining facial images at the local Fog node, where an open-source library facilitates the "zero-registration anti-theft" feature. Beyond identity verification, the system enhances security by uploading captured images to Azure OpenAI to analyze the emotional expressions of individuals, assessing whether they appear nervous to improve the reliability of theft detection. In the event that the system detects a mismatch between the depositor and retriever, it utilizes MQTT messaging to trigger all alarms within the local area network, including buzzers and LCD displays, to alert bystanders and staff immediately. Future expansions of the project could involve adopting more advanced facial recognition models to further increase accuracy and reliability. Additionally, the integration of autonomous AI robotics could be explored to assist in securing premises and intervening during theft alerts, ensuring the ultimate protection of every guest's property.

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